

URL Shortener — Integration Points

Project: URL Shortener

Technology: Python, FastAPI, SQLAlchemy, SQLite, Pydantic

Location: /home/dominic/Documents/fastapiurl

Report #: 17/20

Integration	Type	Risk
SQLite (file)	Local storage	File locking under concurrent access
None	—	Fully self-contained

Integration Points

Internal Integrations

Source	Target	Protocol	Data	Frequency
Auth → Database	SQLAlchemy	User data	Per request	
Chat → OpenAI API	HTTP	Messages	Per request	
Queue → WebSocket	WS	Progress	Real-time	
Frontend → API	HTTP/WS	All	Per action	

External Integrations

| OpenAI | api.openai.com | HTTP/REST | AI chat completions | | GitHub Pages | github.io | HTTPS | Static site hosting | | Render | onrender.com | HTTPS | Backend hosting |

API Integration Details

OpenAI API (AI Chat Proxy)

- **Endpoint:** POST https://api.openai.com/v1/chat/completions
- **Auth:** Bearer token (server-side key)
- **Protocol:** REST + SSE streaming
- **Rate limit:** 60 req/min (proxy level)
- **Models supported:** gpt-4, gpt-3.5-turbo
- **Timeout:** 60 seconds
- **Error handling:** 502 Bad Gateway on failure

GitHub Pages

- **Service:** Static hosting
- **Protocol:** HTTPS + CDN
- **Auto-deploy:** Git push to master triggers rebuild
- **Used by:** WebBartosz, Reports

Integration Health Monitoring

- All services expose `/health` endpoint
- WebBartosz polls every 60 seconds
- Render dashboards show service uptime
- No external monitoring service (e.g., Datadog, Grafana)

Integration Testing

Integration	Test Coverage	Notes
Database		Test with SQLite in-memory
OpenAI		Mocked in CI, live in prod
WebSocket		Tested manually
Auth JWT		Unit tests for token creation